Project Update 4&5

Section 1. Data Collection

The XLK stock price we are looking at is collected by S&P Dow Jones Indices and State Street Global Advisory. S&P Dow Jones Indices is in charge of creating the company benchmark on entering the technology company stock pool; while the State Street Global Advisors selected particular stocks into the combination of the XLK stock. The data we are using is collected from the investing.com website, a financial data platform that provides public real-time market data directly from stock exchanges such as the New York Stock Exchange and NASDAQ. These two trading platforms provide a transparent record of the performance of the Technology Selected Sector Index. We are interested in the influence of the large-cap U.S. technology companies such as Microsoft, NVIDIA, and Apple whose market valuations and returns constitute the movement of the XLK fund.

The variables we are using includes:

- Price (Close): The final price at which the ETF traded when the stock exchange closed (typically 4:00 PM EST).
- Open: The price of the very first transaction when the market opened (9:30 AM EST).
- High & Low: The maximum and minimum prices reached during the trading session.
- Vol.: The total number of shares traded during the day (e.g., 15.90M means 15.9 million shares).
- Change: The percentage difference between the current day's closing price and the previous day's closing price.

The data we are using is collected daily from January 3, 2017, to March 28, 2026, which incorporates the AI outbreak time.

Section 2. Variable Creation and Data Cleaning

We observed outliers that are double the price of the remaining data points. We also found that for most weekends, prices are carried over from the Friday of the same week. These outliers correspond to weekend days, and theey have no data for the variable, Vol. We decided to remove these outliers because they are not consistent with the general pattern, which they doubled the Friday price. Lastly, we split the dataset into two groups using November 30, 2022 (the release date of ChatGPT) as the cutoff.

Section 3. Analytic Methods

EDA

Table 1: Summary Statistics of Stock Price

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mean_price	sd_price	min_price	max_price
69.91683	33.95005	24.4	152.07

Table 1 shows the mean, standard deviation, minimum and maximum of the stock price between 2017/1/3 to 2026/3/27.

Transform Data

Figure 1 shows the stock price over time. The red dashed line indicates the time when ChatGPT was released. Overall, the stock price shows a general upward trend during the period shown.

Figure 2 indicates that the time series is not stationary, as the ACFs are out of the boundary and decay very slowly. Thus, we need to transform the data to make it stationary. For stock data, taking the natural log is a common way to transform the data.

Figure 3 shows the log return over time after log transformation. The log return fluctuates around zero, and the fluctuations are small for most of the time. However, there are a few large spikes. The biggest fluctuations occur around 2020 when the pandemic began. This suggests big fluctuations during pandemic.

Figure 4 shows the ACF of log return. Most ACFs fall within the bound. And Figure 3 suggests the mean is around zero. Thus, the log return time series looks stationary.

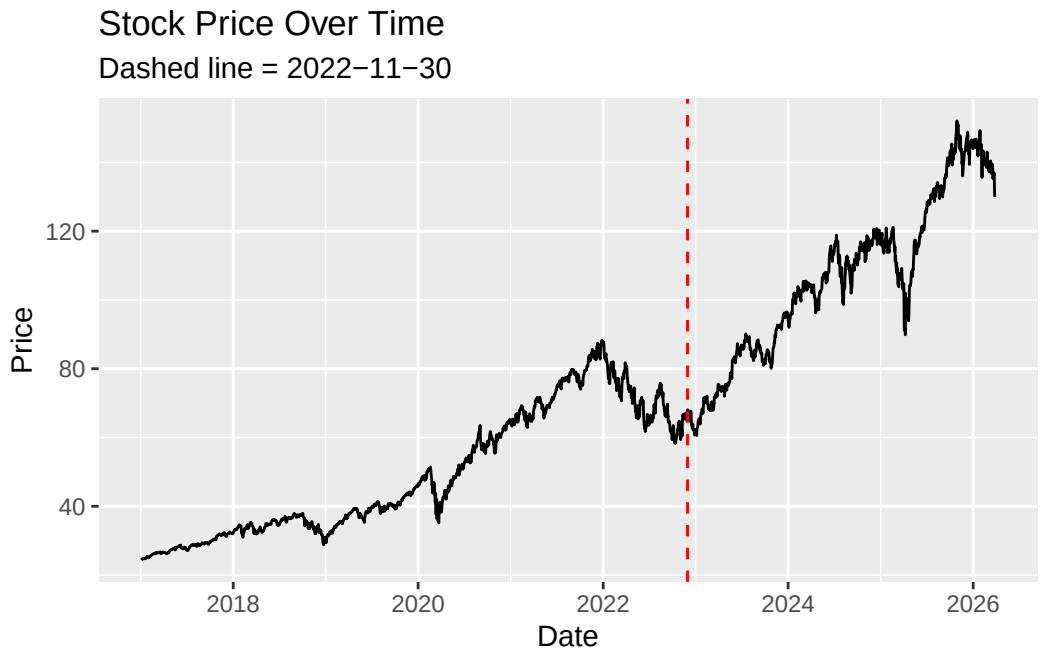

Figure 1

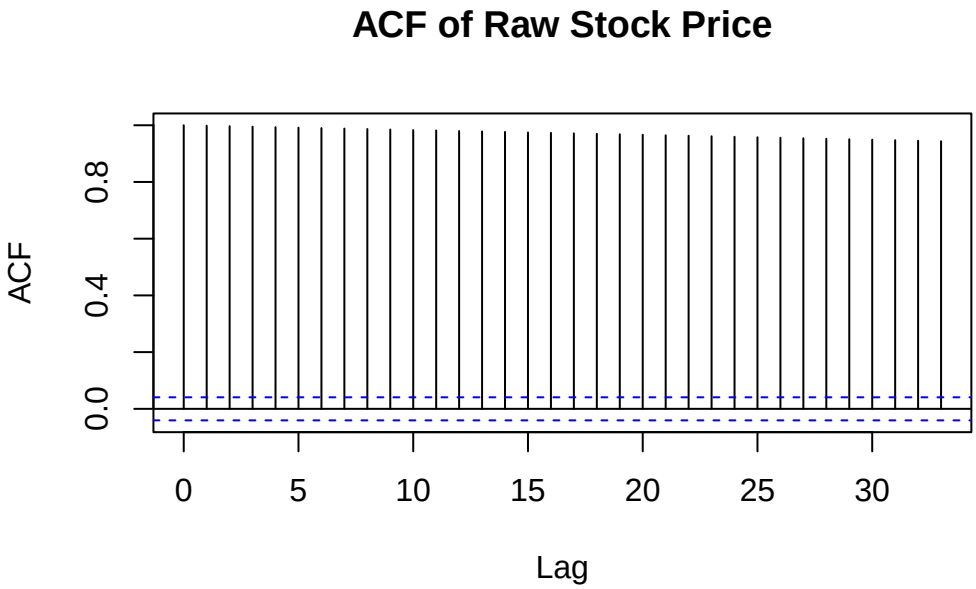

Figure 2

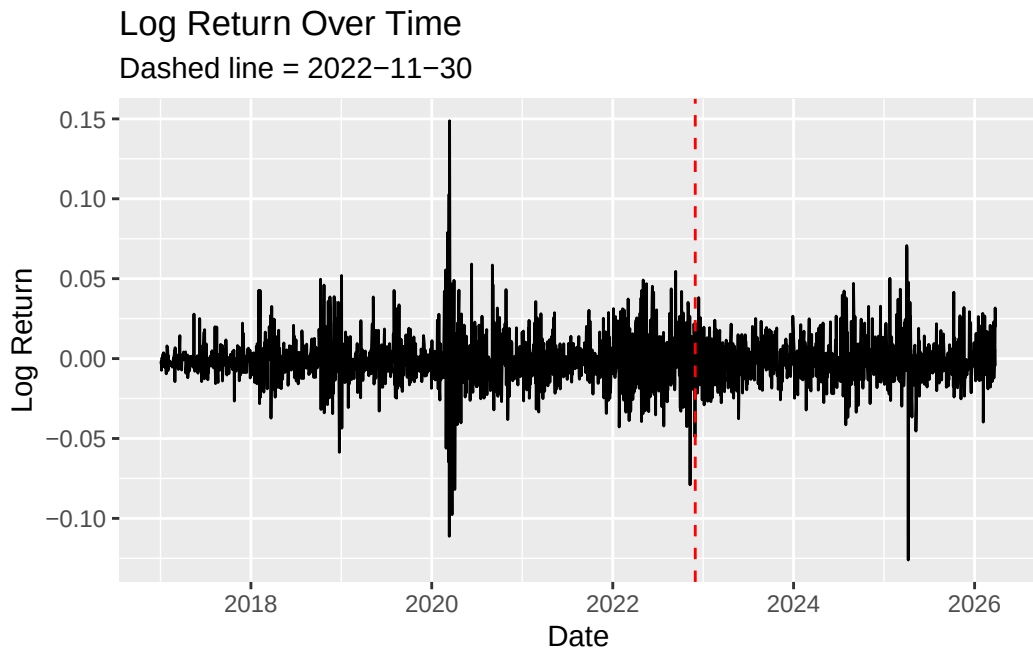

Figure 3

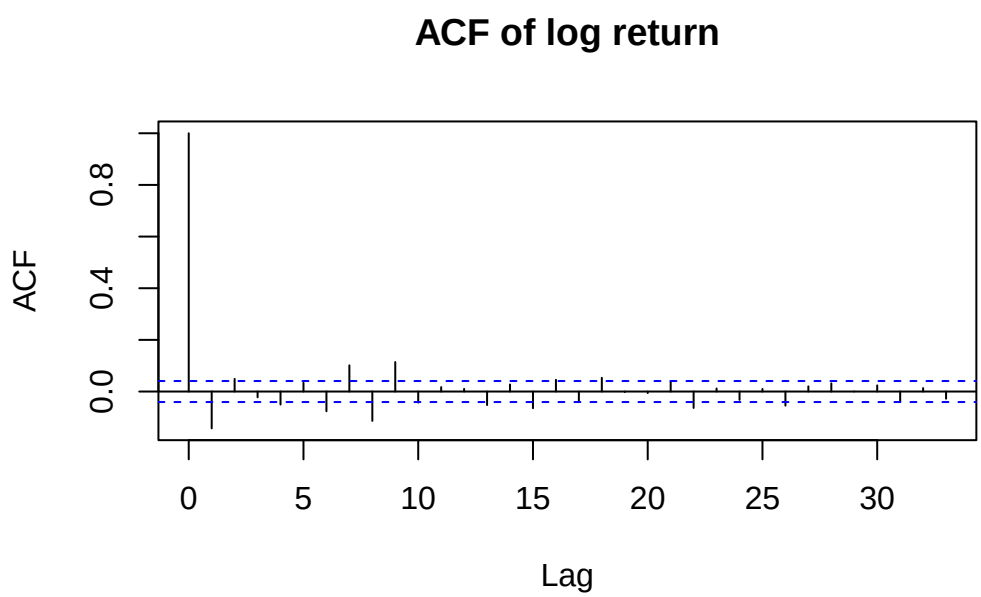

Figure 4

Compare log return

Table 2: Summary Statistics of Stock Price

period	Mean_Return	SD_Return
After	-0.00078	0.01456
Before	-0.00069	0.01641

According to table Table 2, the mean log return before and after the appearance of ChatGPT is both close to zero.

Fit Model

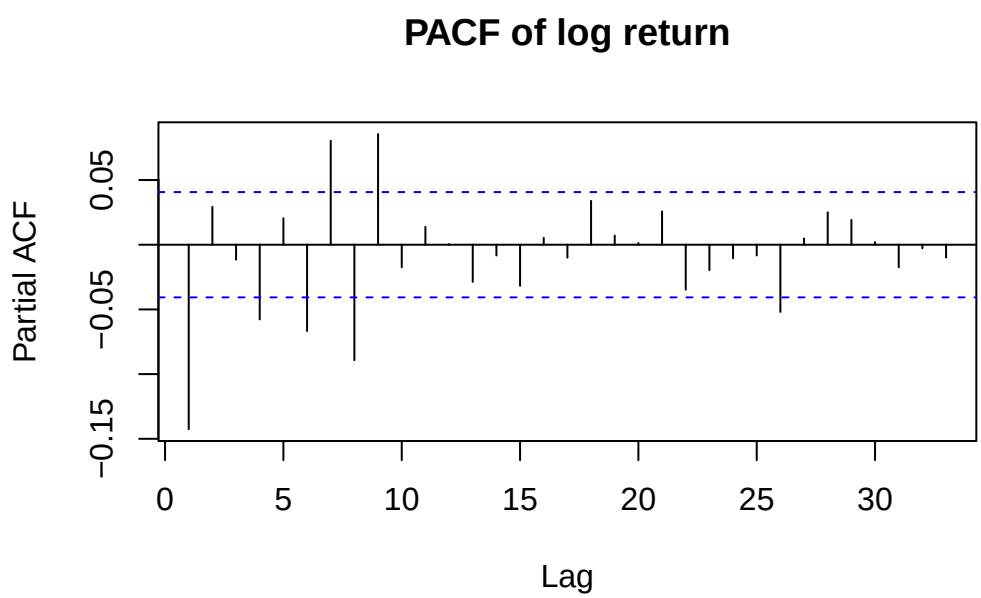

Figure 5

According to Figure 5, there is a strong pike at lag=1. So we fit AR(1), MA(1), and ARMA(1) to capture the time series.

```
Call:
arima(x = data$log_return, order = c(1, 0, 0))
```

```
Coefficients:
          ar1  intercept
      -0.1426     -7e-04
s.e.   0.0206      3e-04

sigma^2 estimated as 0.0002436:  log likelihood = 6359.22,  aic = -12712.45


Call:
arima(x = data$log_return, order = c(0, 0, 1))

Coefficients:
          ma1  intercept
      -0.1337     -7e-04
s.e.   0.0199      3e-04

sigma^2 estimated as 0.0002439:  log likelihood = 6357.63,  aic = -12709.25


Call:
arima(x = data$log_return, order = c(1, 0, 1))

Coefficients:
          ar1     ma1  intercept
      -0.3122  0.1725     -7e-04
s.e.   0.1136  0.1171      3e-04

sigma^2 estimated as 0.0002434:  log likelihood = 6360.17,  aic = -12712.34
```

Model Comparison

```r
# BIC for the models
BIC_AR1<- BIC(AR1)
BIC_MA1<- BIC(MA1)
BIC_ARMA1<- BIC(ARMA1)
```

Table 3: Comparison of AR(1), MA(1), and ARMA(1,1) Models

Model	Coefficients	AIC	BIC	Log Likelihood Ratio
AR(1)	ar1 = -0.1426	-12712.45	-12695.20	6359.225
MA(1)	ma1 = -0.1337	-12709.25	-12692.00	6357.626
ARMA(1,1)	ar1 = -0.3122, ma1 = 0.1725	-12712.34	-12689.35	6360.172

AR(1), MA(1) and ARMA (1,1) are fitted to the log return series. The AR (1) has the smallest AIC among all the three models. The ARMA (1,1) has a slightly higher AIC compares to AR(1), while the MA(1) has the overall largest AIC.

Forcasting

We are interested in how significant the data after 2022/11/30, when ChatGPT got launched, is to forecasting future stock market prices. We will being using data frames of different time sets: 2017/1/3 - 2026/3/27, 2017/1/3 - 2022/11/30 and 2022/11/30 - 2026/3/27. (Date) The forecast models we will be using three of the "benchmark" methods: Mean method, Naive method and Seasonal naive method. (descibe methods) Drift method and decomposition method seem unnecessary with log return data because we have already removed the trend.

First, we aimed to examine whether AI has had an unexpected impact on the stock market and contributed to its fluctuations. We would like to use data from 2017/1/3 - 2022/11/30 to forecast data after 2022/11/30.

Table 4: Forecast Errors

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Method	MAE	RMSE
Mean	0.01049	0.01457
Naive	0.01241	0.01597
Seasonal	0.02161	0.02880

According to Table 4(explain meaning), all the scaled dependent errors are small, even for these easier approaches. It appears that the data prior to 2022/11/30 can effectively predict the data following the cutoff point. Furthermore, this suggests that AI have not yet contribute a significant impact in the stock market. As a result, we will use the entire time frame, allocating 80% for training and 20% for testing, to build a forecast model in predict prices in 2026.

Table 5: Forecast Errors-full model

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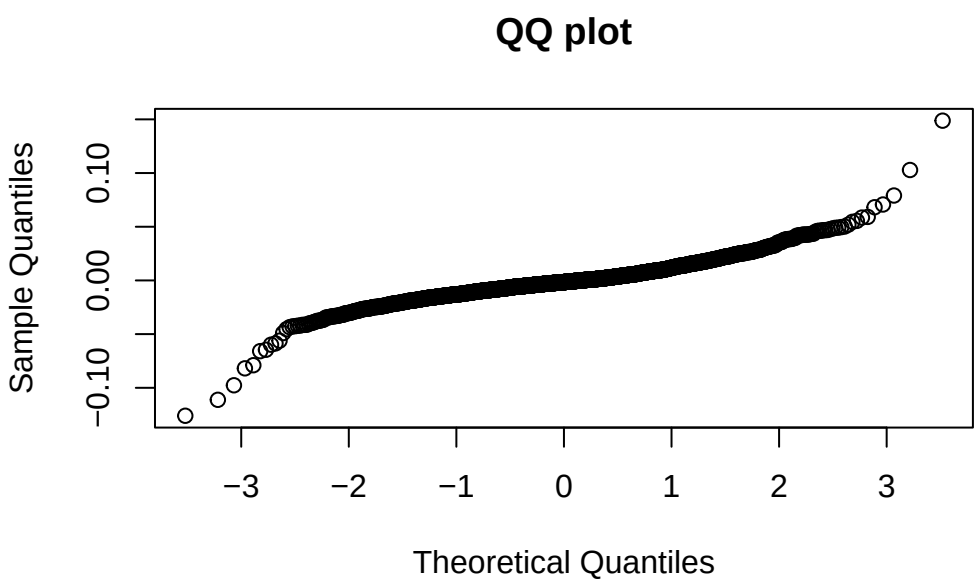
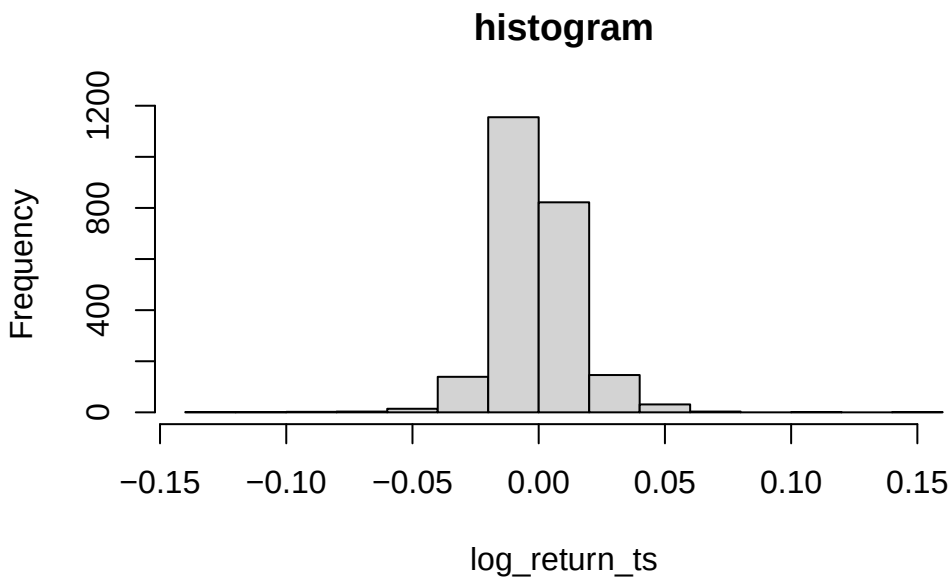
Method	MAE	RMSE
Mean	0.01128	0.01631
Naive	0.01459	0.01845
Seasonal	0.01311	0.01754

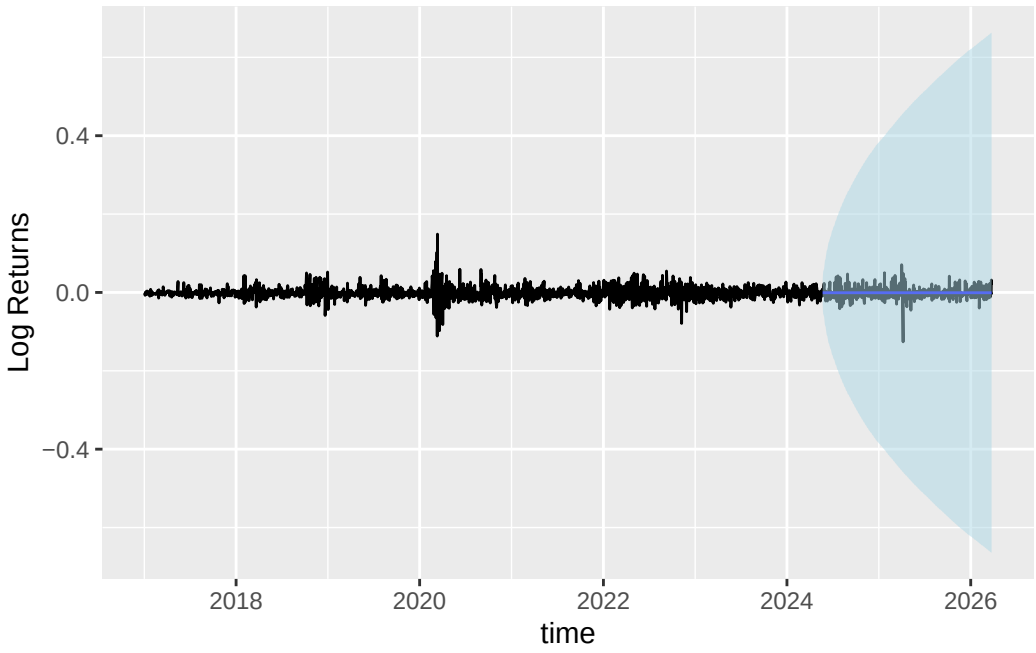
Table 5 shows the mean method is the best with lowest MAE and RMSE. We did not put MAPE here for comparison, because our log return data moves around 0, where some small values here make MAPE enormous and not as a good parameter to look at.





Next, we would like to also make a prediction interval for this forecast model.[ACF:( ]. The histogram looks normal, however qq-plot reveals that log return does not satisfy the linearity assumption. This suggests that our prediction interval is off.

(Will do ARIMA but also keeping this)



Summary

The project test whether the rise of AI has significantly impacted the behavior of technology sector stock returns. Using the XLK ETF price data from 2019 to 2026, we transformed the series into log returns to stationaize the data and applied AR(1), MA(1), and ARMA (1,1) time series models. The result suggests that stock return pattern remained relatively stable befoe and after the emergence of AI, indicating that AI has not yet had a significant impact on the technology stock market price. More broadly, this shows that while AI is often seen as a transformative force, its impact on financial markets may not be immediately significant on stock market return fluctuation.

Limitation

There are some potential limitations for the dataset. One of the limitations is that there's possibility for State Street to rebalance the stock composition over time, thus the data might not be consistent to offer a robust overtime analysis.

Another limitation is the dependency of the dataset. In the EDA, we found out the time series data points are highly correlated whether before or after the data transformation. For instance, the QQ plot indicates deviations from normality, suggesting that model assumptions may not

hold and prediction intervals may be unreliable. In addition, our model only test on the short-term impacts, but AI boom may results from a long-term trend. The log transformation may mask the underlying long term trend.

Also, AR, MA, ARMA are all autoregressive models that look solely on the past values. They do not include other important factors, such as government policy changes, economic policy changes, etc. There are many other factors that affects the stock market that we did not capture their relationship, such as economic indicators and interest rates.

Future Directions

To account for these factors that are not included in our model but would affect the stock market, we could add these time series as predictors in our model. This can help our model better estimate the returns or prices.

We have not been able to conclude AI has significant impact on the behavior of technology sector stock returns. It is still useful to estimate how sensitive technology sector stock price comparing to the other stocks that includes other companies.

It is possible to incorporate the method proposed by Campos-Martins (2024) with our model. After identifying notable development by AI companies, we can refine our model's predictors to improve accuracy.